

Bouira University

Cloud Computing
Part 2

Master: GSI & ISIL

Department of Computer Science.

Presented by
Dr S.BENBATATA

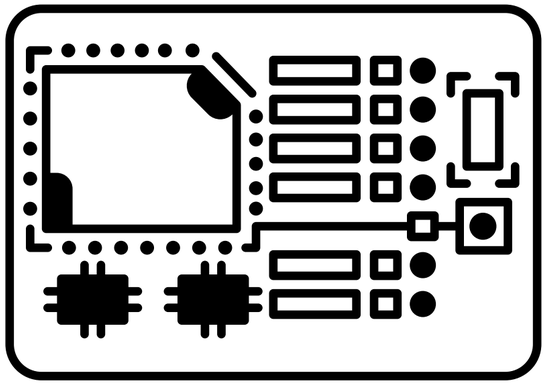
what is visualization?

Virtualization is the core technical foundation upon which all major cloud providers are built including AWS, Google Cloud, Microsoft Azure, IBM, Oracle, and Alibaba Cloud.

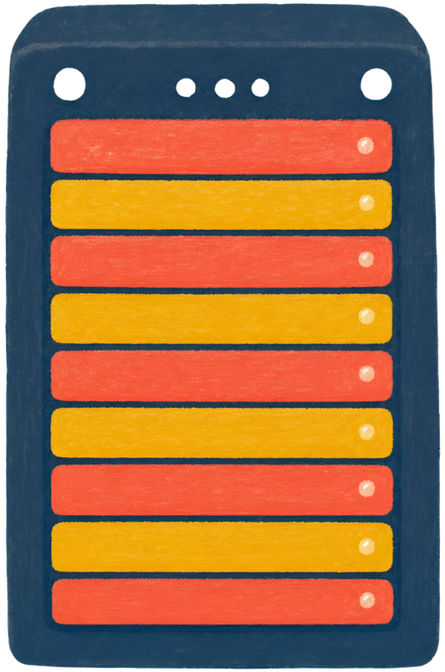
what is visualization?



physical hardware



virtual machine







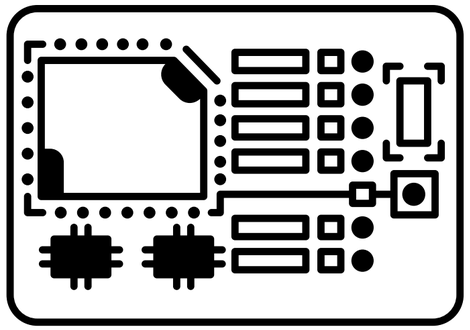




what is visualization?



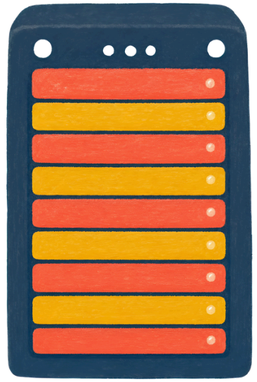
physical hardware



Hypervisor



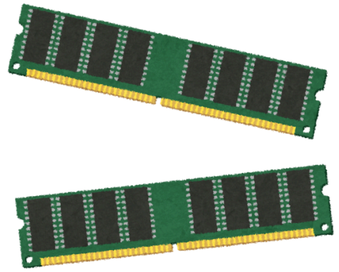
virtual machine



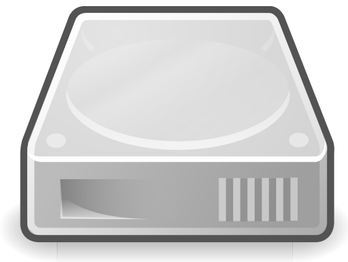


Virtualization is the process of taking a single physical machine (physical hardware) and dividing it into multiple "virtual machines" (VMs).

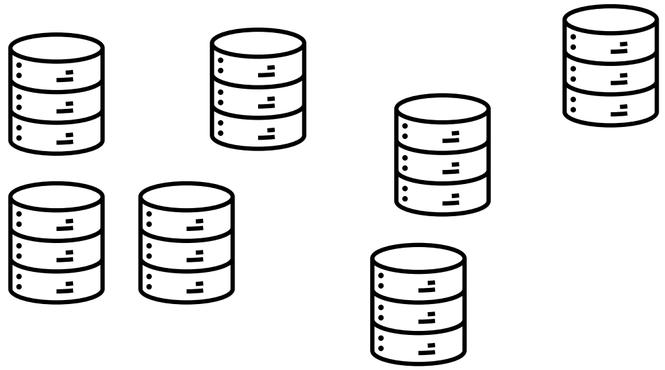
what is visualization?

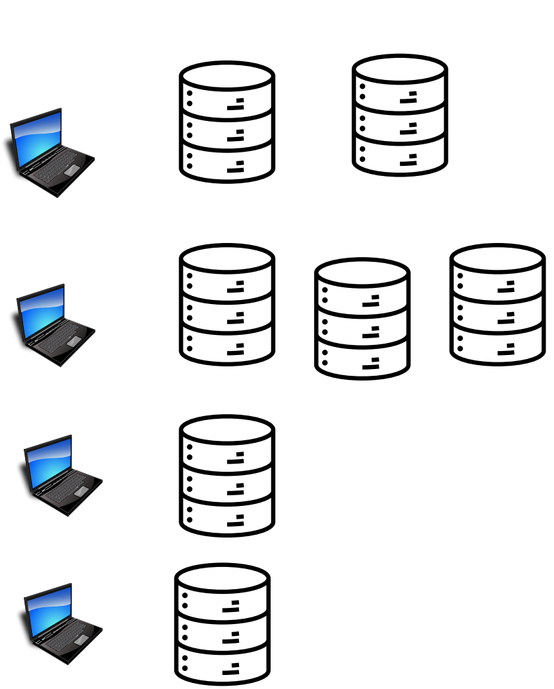


32GB



1TB





8 GB

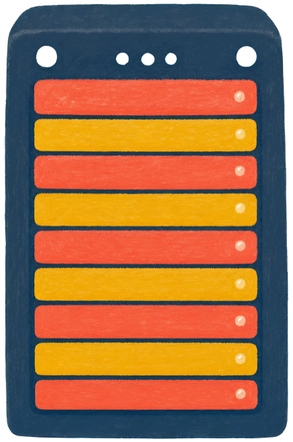
16GB

4GB

4GB

Hypervisor





Architecture

cloud computing is built based on many concepts, including the distributed systems approach proposed by Tanenbaum to provide services over the internet. In addition, virtualization is a core concept of cloud computing. Through virtualization techniques, providers can abstract fundamental computing elements, such as storage, network, hardware, runtime, etc. The internet itself acts as the main medium for the delivery of services.



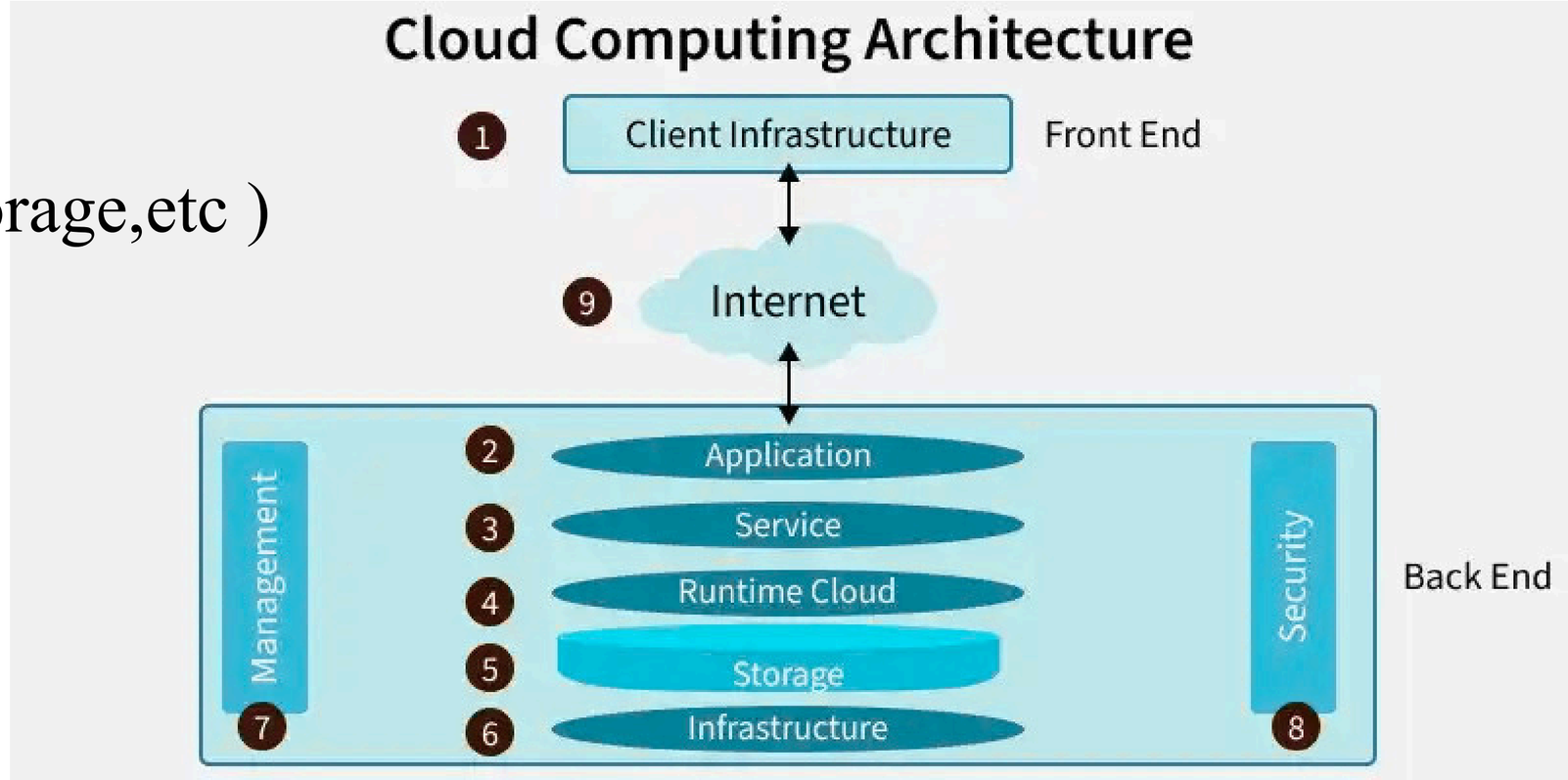
Architecture

The architecture refers to the components and sub-components required for building a model. In cloud computing, architecture defines how these components—such as servers, storage, networks, virtualization layers, and management software—are organized, connected, and interact with one another to deliver services.

Architecture

The architecture of cloud computing may be categorized into three main components:

- Front end (client)
- Back-end (Servers, Storage,etc)
- Network



Architecture

- Front end

Includes user interfaces and client devices. Thin clients use web browsers for simple access, while fat clients provide advanced features and a better user experience.



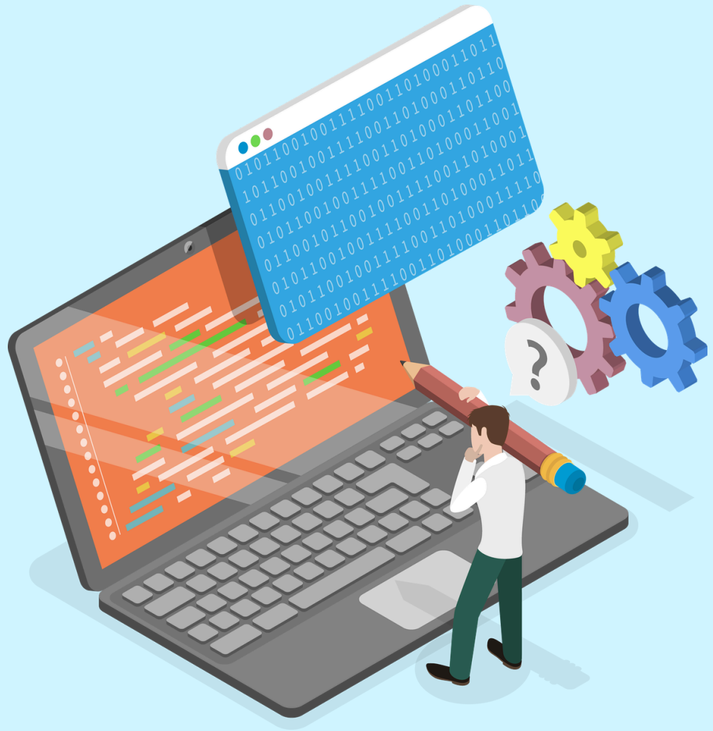


Architecture

- Back-End Platforms

Consists of servers and storage systems that handle data processing, application management, and data storage in the cloud.





Architecture

- Network

  Provides access to cloud services through the Internet.



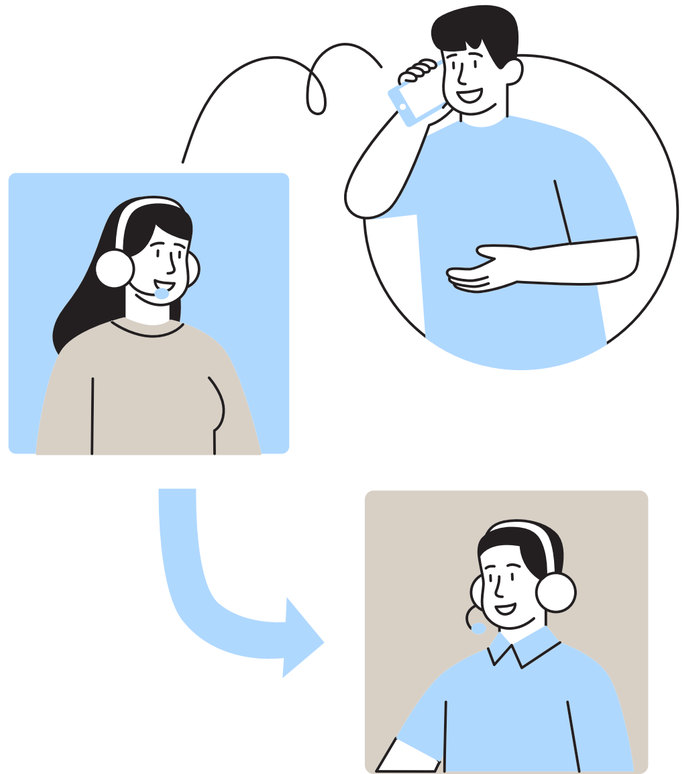
Examples of cloud providers

Examples of cloud providers include:

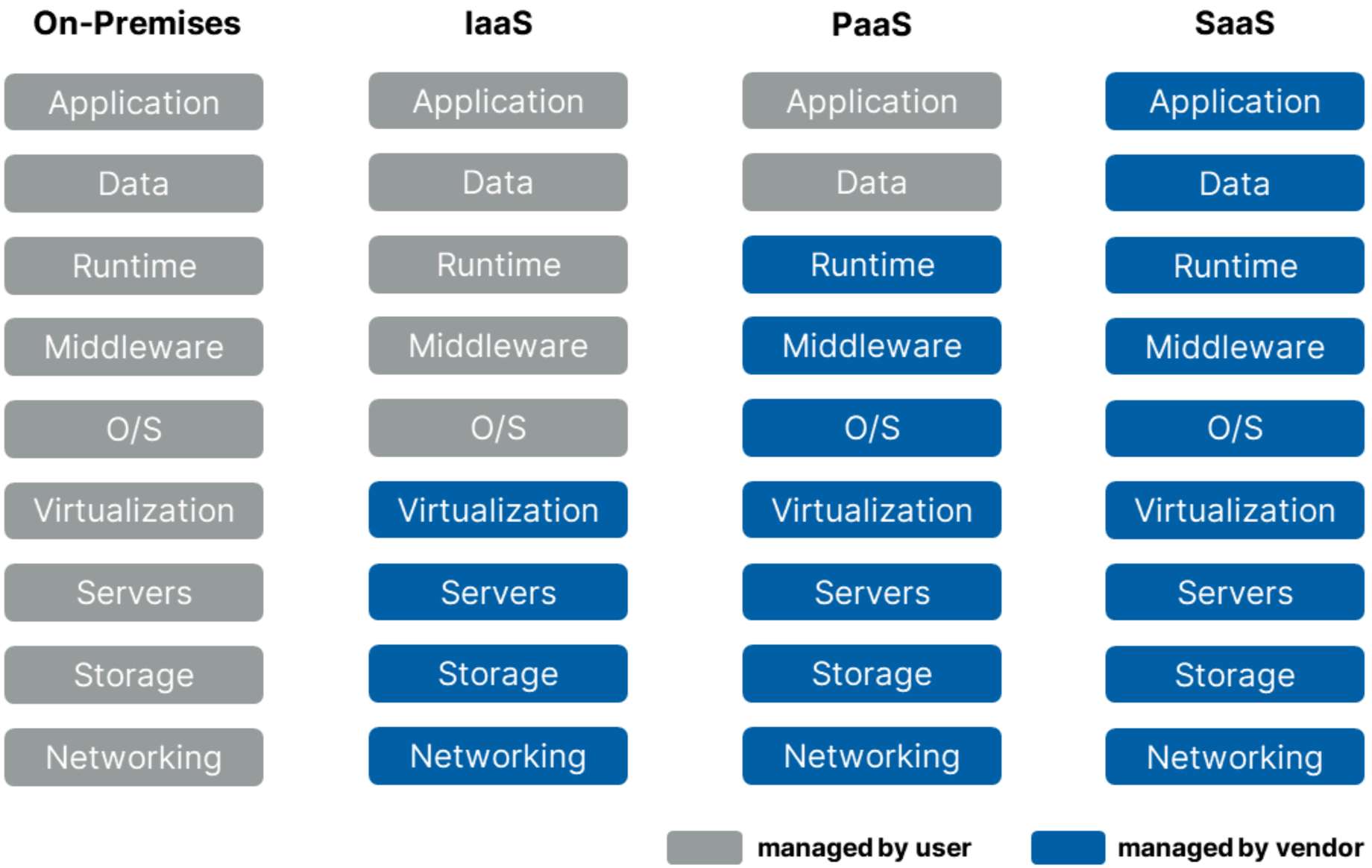
- Amazon Web Services (AWS)
- Google Cloud
- Microsoft Azure
- IBM Cloud
- Alibaba Cloud
- Oracle Cloud
- Salesforce Cloud

Cloud computing models

Cloud computing comes in three different service models Software as a Service (SaaS), Platform as a Service (PaaS) and Infrastructure as a Service (IaaS), each serving a unique set of business needs.



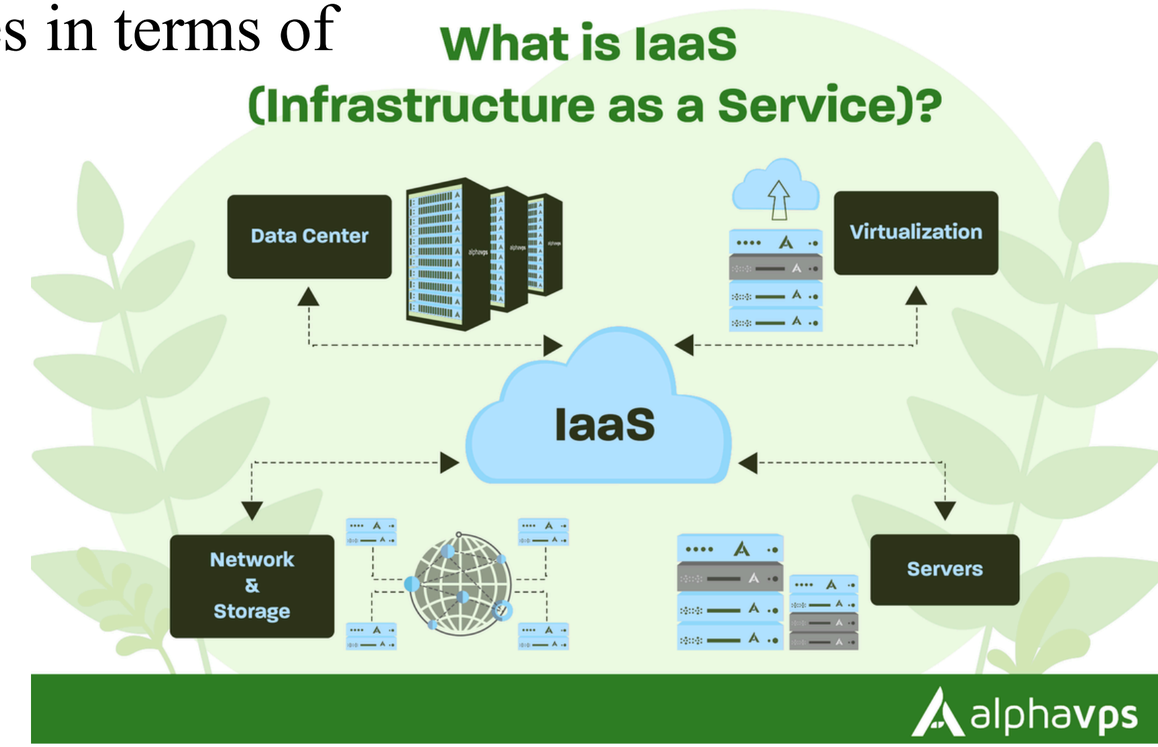
Cloud computing models



Cloud computing models

Infrastructure as a Service (IaaS)

This is achieved by using hardware virtualization techniques, which offer one or more virtual machines, each with its own properties and features in terms of memory size, disk storage, the number of CPUs, and network components, etc.



Cloud computing models

Infrastructure as a Service (IaaS)

Example: Amazon EC2 is an Infrastructure as a Service (IaaS) that provides virtual servers on the cloud, allowing users to create, manage, and scale computing resources as needed.

Benefits of using IaaS:

- Flexibility and Control: It comes up with providing virtualized computing resources such as VMs, Storage, and networks facilitating users with control over the Operating system and applications.
- Reducing Expenses of Hardware: It provides business cost savings with the elimination of physical infrastructure investments making it cost-effective.
- Scalability of Resources: The cloud provides in scaling of hardware resources up or down as per demand facilitating optimal performance with cost efficiency.

Cloud computing models

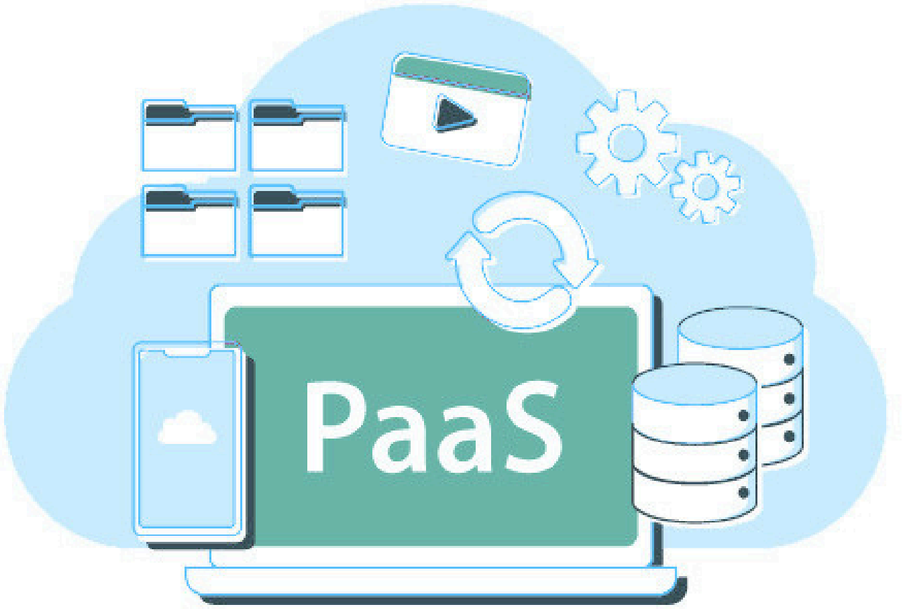
Platform as a Service (PaaS) :

With this model, everything you need to build applications over the Internet, including development tools, infrastructure, and operating systems is provided. This allows developers to work more efficiently by focusing on building and deploying instead of managing applications and dealing with infrastructure. PaaS services include application design, development, and hosting.



Cloud computing models

Platform as a Service (PaaS) :



For example PaaS gives developers a ready made platform to build and deploy apps in the cloud such as :

AWS Elastic Beanstalk is a Platform as a Service (PaaS) by Amazon Web Services that helps developers deploy and manage applications easily while AWS handles servers, scaling, and load balancing, etc.

Some people refer to it as a development platform when talking about the programming layer, where virtualization helps create a runtime environment to build and run applications. Others call it application Platform as a Service (aPaaS) .

Cloud computing models

Platform as a Service (PaaS) :



Benefits of using PAAS:

- Simplifying the Development: It offers application development by keeping the underlying Infrastructure as an Abstraction. It helps the developers to completely focus on application code and background operations are completely managed by the AWS platform.
- Enhancing Efficiency and Productivity.
- Automation of Scaling.

Cloud computing models

Software as a Service (SaaS)

SaaS is a way of using software over the internet instead of installing it on your computer. The software is hosted by a company, and you can use it just by logging in through a web browser. You don’t need to worry about updates, maintenance, or storage the provider takes care of all that.



Cloud computing models

Software as a Service (SaaS)

Example:

Google Docs allows users to create, edit, and share documents online without installing any software.

Canva allows users to create, edit, and share presentations, images, etc., online without installing any software.



Cloud computing models

Software as a Service (SaaS)

Benefits of using SaaS

- Collaboration And Accessibility: It helps users to easily access applications without having the requirement of local installations.
- Automation of Updates: It providers manage the handling of software maintenance .
- Cost Efficiency: It acts as a cost-effective solution by reducing the overhead of IT support by eliminating the need for individual software licenses



Cloud computing models

Everything as a Service (XaaS)

In cloud computing, we have the concept of Everything as a Service (XaaS). This means that if something can be delivered over the internet, it becomes a service in the cloud. In Everything as a Service (XaaS), various services and technologies are provided to users by the cloud, including Infrastructure as a Service (IaaS), Software as a Service (SaaS), and Platform as a Service (PaaS), Function as a Service (FaaS), as we mentioned before about these types. In addition to these, there are other services such as Network as a Service (NaaS), Desktop as a Service (DaaS), Database as a Service (DBaaS), Communication as a Service (CaaS), Healthcare as a Service (HaaS), etc.

Users simply rent or subscribe to access these services provided by the cloud.

Cloud Deployment Model

Cloud Deployment Model is simple a separation which describes where are the company resources deployed.

Cloud Provider



&

Own Datacenter



Cloud Deployment Model

Cloud Deployment Model is simple a separation which describes where are the company resources deployed.

- Public cloud
- Private cloud
- Hybrid cloud

Cloud Provider



&

Own Datacenter



Cloud Deployment Model

Public Cloud

This cloud infrastructure is owned and operated by a third-party cloud service provider (like AWS or Google) and is made available to the general public over the internet. It offers massive scalability and a pay-as-you-go model.

Cloud Provider



Key Characteristics:

- Everything runs on cloud provider hardware
- No local hardware for users.
- Some services share hardware with other customers(multi-tenancy)

Own Datacenter



Cloud Deployment Model

Public Cloud

In a Public Cloud, we find different types of services including **infrastructure, software, and platforms.** The most popular and well known providers in these types, such as: Google AppEngine (PaaS), Amazon EC2 (IaaS), and Salesforce, provide software services in public mode.

In addition, distributed data centers host these services in different geographic locations to ensure high availability and performance. Users can choose the nearest data center to maximize benefits.

Cloud Deployment Model

Public Cloud

Disadvantages

- Not all security and compliance policies can be met.
- No ownership over the physical infrastructure.
- some specific scenarios can’t be done.

Cloud Deployment Model

Private Cloud

Known as « on-premises », is a server, data center, or distributed network that is entirely dedicated to an organization and provides access to systems, services and users within the organization. It’s more secure and safer than the public cloud.

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Key Characteristics:

- Everything runs on your own datacenter
- Self-service should be provided
- You maintain the hardware

Cloud Provider



Own Datacenter



Cloud Deployment Model

Private Cloud

Advantages

- Can support any scenario
- Total control infrastructure
- It offers the highest level of security and control.

Cloud Deployment Model

Private Cloud

Disadvantages

- Initial investment is required (CapEx).
- Very dependent on IT skills & expertise.
- less scalable compared to the public cloud.
- less flexibility.

Cloud Deployment Model

Hybrid Cloud

This model combines a private cloud with one or more public clouds, allowing data and applications to be shared between them.

The hybrid cloud is essentially a heterogeneous distributed system that integrates services from both private and public clouds, leading some researchers to refer to it as heterogeneous clouds

Cloud Provider



Own Datacenter



Cloud Deployment Model

Hybrid Cloud

Hybrid cloud represents the optimal solution, combining the strengths of both public and private models to deliver high performance while addressing diverse user requirements.

Cloud Provider



Own Datacenter



Cloud Deployment Model

Hybrid Cloud

Advantages

- Great flexibility
- You can run any apps in private cloud
- Can utilize existing infrastructure
- offering robust security.
- on-demand scalability.
- cost efficiency
- Can take advantage of all public cloud benefits

Cloud Deployment Model

Hybrid Cloud

Disadvantages

- Initial investment is required (CapEx)
- Very dependent on IT skills & expertise

Cloud Deployment Model

Below table presents high level deployment model separation

Layer	Cloud Provider	Own Datacenter
Public	✅	✖️
Hybrid	✅	✅
Private	✖️	✅

https://marczak.io/az-900/#ep06

Cloud Deployment Model

There are other deployment models besides these main models, such as multi-cloud and community cloud.

Cloud Deployment Model

Community cloud

Community cloud is a distributed model where infrastructure is shared among multiple organizations and users with common interests and goals, such as security, requirements, and compliance. Community cloud models bring several key benefits, including shared costs, enhanced security, easier collaboration, and privacy controls. They also allow organizations with similar goals to stay aligned on compliance requirements

Cloud Deployment Model

Multi-Cloud

It refers to the use of two or more public or private cloud services from different providers within a single architecture. By integrating multiple clouds, organizations can maximize their respective advantages within one architecture.

For example an organization might choose one provider for cost efficiency and another for better management and security.

This approach enables the selection of the most suitable services for specific requirements, while also avoiding vendor lock-in and improving performance and flexibility.